MILITARY SPECIFICATION

CONTROL BOX, COMPUTER LOGIC UNIT,
TEST SET, C4020 GSM-70: 8623274

1. SCOPE

1.1 This specification covers one type of control box used to provide manual entry of commands to the Computer Logic Unit: 8623272.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel: General Specification Governing the Manufacturing and Inspection of |
| MIL-P-14232 | Parts, Equipment and Tools for Ordnance Materiel, Packaging of |
| MIL-I-45607 | Inspection Equipment, Supply and Maintenance of |
| MIL-C-46357 | Computer, Gun Direction: M18; Manufacturing Processes and Inspection of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-108 | Definitions of and Basic Requirements for Enclosures for Electrical and Electronic Equipment |
| MIL-STD-109 | Quality Assurance Terms and Definitions |

DRAWINGS

U.S. Army, Frankford Arsenal

| | |
|---|---|
| D8623498 | Schematic Diagram for Control Box, Computer Logic Unit, Test Set, C4020, GSM-70 |
| F8623274 | Control Box, Computer Logic Unit, Test Set, C4020, GSM-70 |

FSC 1220

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PACKAGING DATA SHEET

8623274 — Packaging of Control Box, Computer Logic Unit, Test Set, C4020, GSM-70

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The control box shall be manufactured in accordance with Drawing F862374 and drawings pertaining thereto.

3.2 General specifications.- The control box shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926.

a. Dimensions and tolerances.
b. Inorganic protective surface finishes.
c. Workmanship.

3.2.2 MIL-C-46357.

a. Wiring.
b. Etched circuit assemblies.
c. Marking.
d. Mounting of components.
e. Soldering process.

3.3 Environmental.

3.3.1 Vibration.- The control box with cover in place shall be capable of withstanding a simple harmonic motion having an amplitude of 0.015 inch (0.030 total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return in approximately one (1) minute. The vibration shall be applied to the unit for a period of 10 minutes in each of the three (3) mutually perpendicular axes (total of 30 minutes). Subsequent to vibration, the unit shall not be damaged or adversely affected and shall meet the requirements of 3.5.

3.3.2 Impact.- The control box, with cover in place shall be capable of withstanding the impact produced by dropping the unit on a solid concrete base from a height of 30 inches. The unit shall meet the requirements of 3.5 after each drop outlined below.

a. One drop on any one corner at ambient room temperature (60° - 90°F).
b. One drop on the diagonally opposite corner immediately after exposure to a temperature of -40°F for one hour.

3.3.3 Temperatures.



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3.3.3.1 Operating.- The control box, with cover removed, shall meet the requirements of 3.5 while thermally stabilized at temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The control box, with cover removed, shall meet the requirements of 3.5 at a temperature between 60°F and 90°F after having been exposed to temperatures of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The control box, with cover removed, shall meet the requirements of 3.5 while exposed to an atmosphere with a relative humidity (RH) up to 95% at a temperature of +135°F.

3.3.4.2 Non-operating.- The control box, with cover removed, shall meet the requirements of 3.5 after exposure to (when thoroughly dried) an atmosphere with an RH of 100% including condensation, for a period of 4 hours.

3.3.5 Immersion.- The control box, with cover in place, shall show no evidence of leakage while immersed under a 3 foot head of water for a period of not less than 2 hours. The control box with cover removed shall meet the requirements of 3.5.

3.4 Electrical.

3.4.1 Wiring.- Wiring, terminals and connections shall be in accordance with Electrical Schematic Drawing D8623498.

3.5 Performance.- Unless otherwise specified, the control box, with cover removed, shall meet the performance requirements at an ambient temperature between 60°F and 90°F.

3.5.1 Neon lamps.- The neon lamps shall light when energized by a plus 150 ± 5% volt direct current (dc) source.

3.5.2 Switches.- The Switches S1 through S16 shall indicate continuity when in normally closed position.

3.5.3 Clock pulse.- The output wave form of the single pulse module shall be as indicated in Figure 1 when energized by a minus 12vdc applied to J1-T and triggered by static-clock switch.



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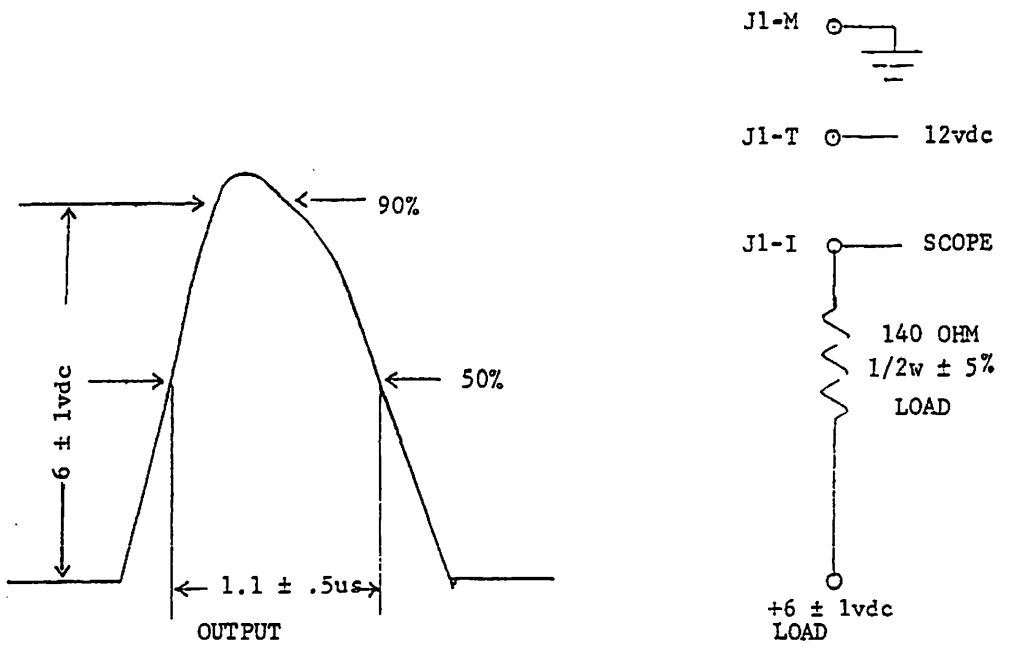



FIGURE 1

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission.

An initial production sample shall be submitted to a Government approved facility for test and evaluation in accordance with the contract schedule. The sample shall consist of two control boxes. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved



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facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- Three control boxes shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. The sample shall have successfully met all other examinations and acceptance tests specified herein, and shall meet the following requirements and tests:

TABLE I

| CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|
| 401. Fabrication | 3.1 | Per applicable dwgs and specs. - visual |
| 402. General specification | 3.2 | MIL-F-13926 and MIL-C-46357 |
| 403. Impact | 3.3.2 | 4.6.1.1 |
| 404. Operating temperatures | 3.3.3.1 | 4.6.1.2 |
| 405. Storage temperatures | 3.3.3.2 | 4.6.1.3 |
| 406. Humidity (operating) | 3.3.4.1 | 4.6.1.4 |
| 407. Humidity (non-operating) | 3.3.4.2 | 4.6.1.4.1 |
| 408. Immersion | 3.3.5 | 4.6.1.5 |

4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.2).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.



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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Environmental.- Testing equipment utilized in the subsequent environmental tests shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 at the test conditions of 3.3.

4.6.1.1 Impact.- The control box shall be dropped as specified in 3.3.2a. Immediately after the impact, the control box shall be examined by visual and tactile means for any physical damage then tested as outlined in 4.6.2 for compliance with 3.3.2.

a. The control box shall be placed into a cold chamber and exposed to a temperature of -40°F for one hour, then dropped as specified in 3.3.2b. Immediately after the impact the control box shall be examined and tested as outlined above.

4.6.1.2 Operating temperatures.- The control box shall be exposed to each temperature extreme specified in 3.3.3.1 for a minimum of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the control box shall be subjected to the tests specified in 4.6.2 for compliance with 3.3.3.1.

4.6.1.3 Storage temperatures.- Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.3.2. The control box shall be exposed to each specified temperature for a minimum of 5 hours. Subsequent to the temperature exposures, the control box shall be stabilized in an ambient temperature between +60° and +90° Fahrenheit. The control box shall show no evidence of damage. Upon successful completion of this examination, the control box shall be subjected to the tests specified in 4.6.2 for compliance with 3.3.3.2.

4.6.1.4 Humidity-operating.- This test shall be performed in accordance with Procedure I of MIL-F-13926, except the relative humidity and



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temperature shall be as specified in 3.3.4.1. While exposed to the specified conditions for one 12-hour cycle, the control box shall be tested as specified in 4.6.2 for compliance with 3.3.4.1.

4.6.1.4.1 Humidity non-operating.- This test shall be performed utilizing the facilities specified above. Upon completion of the 4 hour period, the control box shall be thoroughly dried and then tested as specified in 4.6.2 for compliance with 3.3.4.2.

4.6.1.5 Immersion.- This test shall be performed in accordance with the watertight test specified in MIL-STD-108. Upon completion of this test, the control box shall be thoroughly dried. The cover and panel shall be removed and the interior shall be visually examined for any evidence of damage due to water seepage, then tested as specified in 4.6.2 for compliance with 3.3.5.

4.6.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be performed on a 100% basis. Test conditions shall be as specified in 3.4.

4.6.2.1 Neon lamps.- Connect a +150 ± 5 volt dc supply to the receptacle pin J1-U. The neon lamps listed in Column A of Table II shall light when the respective pins specified in Column B are grounded in the order shown.

TABLE II

| Column A | Column B |
|---|---|
| R1 | e |
| R2 | f |
| R3 | j |
| R4 | k |
| R5 | m |
| R6 | n |
| R7 | p |
| R8 | q |
| R9 | r |

Disconnect the power supply.

4.6.2.2 Switches R1 through R9.- Connect ground to receptacle pin J1-M. Connect a +6 ± 1 volt dc through a 6 volt (min. 0.2 ampere) incandescent lamp to the pins specified in Table V, Column A one at a time. Each lamp shall light. When each switch specified in Column B is depressed, the respective lamp shall extinguish.



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TABLE III

| Column A | Column B |
|---|---|
| A | R1 |
| B | R2 |
| C | R3 |
| D | R4 |
| G | R6 |
| H | R7 |
| J | R8 |
| K | R9 |

Connect the same lamp and dc supply to pin J1-E and ground pin J1-F.

The lamp shall light. Depress Switch R5 and the light shall extinguish. Disconnect the power supply and lamp.

4.6.2.2.1 Switch (clear-operate).- Connect the same power supply and lamp to pin J1-P and ground pin J1-M. The lamp shall light. Place the clear-operate switch to the normally open position and the lamp shall extinguish.

a. Connect the power supply and lamp to pin J1-R and ground pin J1-T. The lamp shall light. Depress the switch and the lamp shall extinguish.

b. Connect the power supply and lamp to pin J1-S and ground pin J1-T. Depress the switch and the lamp shall light. Release and the lamp shall extinguish. Disconnect the power supply and lamp.

4.6.2.2.2 Switches (command).- Connect the same power supply and lamp to J1-d and ground J1-M. The lamp shall light. Depress 1SF, ITF, and ITA and the lamp shall extinguish. Repeat this test with the lamp connected to the pin specified in Column A of Table IV. The lamp shall extinguish when each switch specified in Column B is depressed in the order shown.

TABLE IV

| Column A | Column B |
|---|---|
| c | OSF - ISF - ITF - ITA |
| b | ISA |
| a | OSA - OTF - OTA |
| z | ITF - ITA |
| y | OTF - OTA |
| x | ITA |
| w | OTA |
| v | BSC (either way) |
| l | -18 VFT OFF |
| s | - 18 VFT ON |

Disconnect the power supply and lamp.



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4.6.2.3 Clock pulse.- Connect a -12 ± 1vdc to J1-T and ground J1-M. Connect an oscilloscope to J1-I with the load specified in 3.5.3 to a +6 ± 1vdc. Place clock switch to the clock position. A pulse similar to the pulse shown in Figure 1 shall appear on the oscilloscope. Remove all connections.

4.7 Packaging, packing, and marking.- Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet 8623274. The level of protection shall be as specified in the procurement documents.

6. NOTES

6.1 Intended use.- The control box, computer logic unit is used to provide manual entry of commands normally programmed on tape.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable stock number.
c. Selection of applicable levels of preservation, packaging, and packing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A289Y



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